Data versus Information Sales under Financial Constraints

Dmitry Orlov (UW-Madison)

Andrzej Skrzypacz (Stanford)

Pavel Zryumov (Rochester)

SFS Cavalcade

May 20, 2026

Data and Information Driven Economy

- Firms often have data that can benefit other market participants.
  - Example 1: viewer preference data generated by streaming platforms.
  - Example 2: driving data generated by Uber, Waymo, Tesla used to train self-driving algorithms.
- These data are often quite complex
  - Its quality cannot be verified by a third party, e.g., court
- To what extent can firms produce, share, and sell data efficiently?

Data and Information Driven Economy

- Firms often have data that can benefit other market participants.
  - Example 1: viewer preference data generated by streaming platforms.
  - Example 2: driving data generated by Uber, Waymo, Tesla used to train self-driving algorithms.
- These data are often quite complex
  - Its quality cannot be verified by a third party, e.g., court
- To what extent can firms produce, share, and sell data efficiently?
- Many papers focus on the non-rival properties of information.
  - Jones and Tonetti (2020), Farboodi and Veldkamp (2025).
- This paper studies the implications of the uncertainty of information content for information trade.

Preview of the Paper

- Study information sharing without dynamic commitment.
  - Seller cannot commit how much information to share in the future.
  - Buyer cannot commit to future payments.

- Information is special: its content is inherently uncertain and the resolution of this uncertainty poses risks for efficient communication.

Preview of the Paper

- Study information sharing without dynamic commitment.
  - Seller cannot commit how much information to share in the future.
  - Buyer cannot commit to future payments.

- Information is special: its content is inherently uncertain and the resolution of this uncertainty poses risks for efficient communication.

- Both **dynamic communication** and **information design** are needed to achieve efficiency.
  - Uncertainty in signals implies that information and data may behave differently from capital goods.

Preview of the Paper

- Study information sharing without dynamic commitment.
  - Seller cannot commit how much information to share in the future.
  - Buyer cannot commit to future payments.

- Information is special: its content is inherently uncertain and the resolution of this uncertainty poses risks for efficient communication.

- Both **dynamic communication** and **information design** are needed to achieve efficiency.
  - Uncertainty in signals implies that information and data may behave differently from capital goods.

- Sharing granular signals exposes parties to inefficient risk due to the short-term uncertainty in data content.
  - Efficiency can never be achieved for discrete fundamentals.
  - Slack in the budget is necessary for any non-Gaussian fundamentals.

Related Literature

- **Gradualism**: Che and Sàcovic (2004), Compte and Jehiel (2004), Horner and Skrzypacz (2016), Pitchford and Snyder (1994), Lockwood and Thomas (2002), Admati and Perry (1991).

- **Dynamic Bayesian Persuasion**: Ely (2017), Ely and Szydlowski (2020), Orlov, Skrzypacz and Zryumov (2020), Brocas and Carillo (2007).

- **Multiple Receivers**: Rosenberg, Solan, and Vieille (2013), Kolotilin, Mylovanov, Zepechelnyuk and Li (2017), Orlov, Zryumov and Skrzypacz (2023), Inostroza (2019).

Model

Players and Preferences

- Time is continuous $t \in [0, 1]$ (keeps track of sequence of events).
- Fundamental **state** $\theta \in \{0, 1\}$, with common prior $p_0 = \mathrm{P}(\theta = 1)$.
- Two players: Sender and Receiver.
- Quasi-linear preferences over **terminal** beliefs p and money m:

$$\boldsymbol{U_i(p, m) = u_i(p) + m} \qquad i \in \{R, S\}.$$

  - Receiver likes information sharing: $u_R(p)$ is convex.
  - Sender dislikes information sharing: $u_S(p)$ is concave, e.g., competition.
- Gains from information ("trade"): $u_R(p) + u_S(p)$ is convex in p.
  - Full information sharing is efficient.
- Receiver has a starting budget $\bar{M}$, Sender is penniless.
  - Can the Receiver pay to induce efficient information sharing?

Information Flow and Timing

- Sender has access to a Gaussian d.g.p. $X = (X_i)_{i=0}^{\infty}$ about θ:

$$dX_i = \theta\, di + \sigma\, dB_i.$$

- Each **"data"** unit $\boldsymbol{dX_i}$ is an **infinitesimal granular signal** about θ.
  - Observing the entire data-stream fully reveals θ.
  - Infinitesimal signals are divisible $\Rightarrow$ no built-in need to split signals.

Information Flow and Timing

- Sender has access to a Gaussian d.g.p. $X = (X_i)_{i=0}^{\infty}$ about θ:

$$dX_i = \theta\, di + \sigma\, dB_i.$$

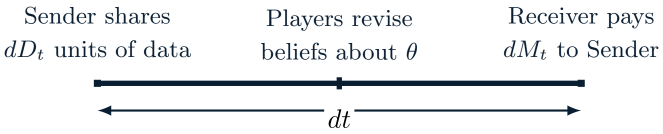
- Each **“data”** unit $\boldsymbol{dX_i}$ is an **infinitesimal granular signal** about θ.
- $\boldsymbol{D_t}$: quantity of data shared by Sender up to time t.
  - Sender discloses process X to Receiver over time.
  - Both players observe X_i for $i \in [0, D_t]$.
- $\boldsymbol{M_t}$: cumulative payment to the Sender up to time t.
  - Receiver makes payments to Sender over time.
  - Payments are irreversible, e.g., capital investments.

Information Flow and Timing

- Sender has access to a Gaussian d.g.p. $X = (X_i)_{i=0}^{\infty}$ about θ:

$$dX_i = \theta\, di + \sigma\, dB_i.$$

- Each **“data”** unit $\boldsymbol{dX_i}$ is an **infinitesimal granular signal** about θ.
- $\boldsymbol{D_t}$: quantity of data shared by Sender up to time t.
  - Sender discloses process X to Receiver over time.
  - Both players observe X_i for $i \in [0, D_t]$.
- $\boldsymbol{M_t}$: cumulative payment to the Sender up to time t.
  - Receiver makes payments to Sender over time.
  - Payments are irreversible, e.g., capital investments.



Data Sharing Strategy

- D_t: quantity of data shared by Sender up to time t.
  - Both players observe X_i for $i \in [0, D_t]$.

Definition

Data-sharing strategy is a weakly increasing **continuous** process $D = (D_t)_{t\in[0,1)}$ that is **predictable** w.r.t. the induced filtration $\mathbb{F}$.

Data Sharing Strategy

- D_t: quantity of data shared by Sender up to time t.
  - Both players observe X_i for $i \in [0, D_t]$.

Definition

Data-sharing strategy is a weakly increasing **continuous** process $D = (D_t)_{t\in[0,1)}$ that is **predictable** w.r.t. the induced filtration $\mathbb{F}$.

- **Predictability**: Sender decides how much data to share in advance
  - The amount of data shared at time t should be known based on the information available at $t-$.
  - Discrete time: every period t choose how many observations X_i to share based on history $t-1$

Data Sharing Strategy

- D_t: quantity of data shared by Sender up to time t.
  - Both players observe X_i for $i \in [0, D_t]$.

Definition

Data-sharing strategy is a weakly increasing **continuous** process $D = (D_t)_{t\in[0,1)}$ that is **predictable** w.r.t. the induced filtration $\mathbb{F}$.

- **Predictability**: Sender decides how much data to share in advance
  - The amount of data shared at time t should be known based on the information available at $t-$.
  - Discrete time: every period t choose how many observations X_i to share based on history $t-1$
- **Continuity**: Sender observes all "intermediate" data points as they are shared.
  - Cannot package data into blocks.

Information Sharing Strategy

Definition

Information-sharing strategy is a weakly increasing process $D = (D_t)_{t\in[0,1)}$ that is **adapted** to the induced filtration $\mathbb{F}$.

Information Sharing Strategy

Definition

Information-sharing strategy is a weakly increasing process $D = (D_t)_{t \in [0,1)}$ that is **adapted** to the induced filtration $\mathbb{F}$.

- The Sender can disclose a block of data X at time t:
  - D_t jumps at time t;
  - X_i with $i \in (D_{t-}, D_t]$ are shared.

Information Sharing Strategy

Definition

Information-sharing strategy is a weakly increasing process $D = (D_t)_{t\in[0,1)}$ that is **adapted** to the induced filtration $\mathbb{F}$.

- The Sender can disclose a block of data X at time t:
  - D_t jumps at time t;
  - X_i with $i \in (D_{t-}, D_t]$ are shared.
- The "size" of the block can depend on the data contained in the block itself.
  - Intuitively, the Sender can buffer new data. Once the data in the buffer reaches some threshold, the Sender shares the entire buffer as a single block.
  - E.g. $D_t = \inf\{I > D_{t-} : X_I - X_{D_{t-}} > c\}$ for some $c > 0$.

Information Sharing Strategy

Definition

Information-sharing strategy is a weakly increasing process $D = (D_t)_{t\in[0,1)}$ that is **adapted** to the induced filtration $\mathbb{F}$.

- The Sender can disclose a block of data X at time t:
  - D_t jumps at time t;
  - X_i with $i \in (D_{t-}, D_t]$ are shared.
- The "size" of the block can depend on the data contained in the block itself.
  - Intuitively, the Sender can buffer new data. Once the data in the buffer reaches some threshold, the Sender shares the entire buffer as a single block.
  - E.g. $D_t = \inf\{I > D_{t-} : X_I - X_{D_{t-}} > c\}$ for some $c > 0$.
- Lemma: Information-sharing strategies allow for Bayesian Persuasion within period dt.

Hold-up Problem

- **Receiver Hold-up:** if Sender shares all data/information upfront, then Receiver can withhold payments.
- **Sender Hold-up:** if Receiver pays up-front, then Sender can withhold information.

Hold-up Problem

- **Receiver Hold-up:** if Sender shares all data/information upfront, then Receiver can withhold payments.
- **Sender Hold-up:** if Receiver pays up-front, then Sender can withhold information.
- Hold-up problems prohibit efficient communication with static strategies.
  - Even though the optimal efficient outcome is static: communicate all data/information in exchange for individually rational payment.
  - Even though they can design information flexibly, they cannot contract/commit to it.
  - Applies if data is not verifiable by a third party.

Hold-up Problem

- **Receiver Hold-up:** if Sender shares all data/information upfront, then Receiver can withhold payments.
- **Sender Hold-up:** if Receiver pays up-front, then Sender can withhold information.
- Hold-up problems prohibit efficient communication with static strategies.
  - Even though the optimal efficient outcome is static: communicate all data/information in exchange for individually rational payment.
  - Even though they can design information flexibly, they cannot contract/commit to it.
  - Applies if data is not verifiable by a third party.
- **Unique Static Nash** equilibrium = no data sharing and no payments.
  - Static Nash is the worst continuation equilibrium, e.g., punishment.
- Can the players do better if they communicated over several periods?

Equilibrium Definition

Definition

An equilibrium is a pair of the Sender's sharing strategy $D^* = (D_t^*)_{t\in[0,1)}$ and Receiver's payment strategy $M^* = (M_t^*)_{t\in[0,1)}$, that satisfy

- Sender incentive compatibility:

$$\mathrm{E}\Big[\underbrace{u_S(p_1^*) + M_1^*}_{\textit{Sender equilibrium payoff}} \,\Big|\, \mathcal{F}_{t-}\Big] \quad \geq \quad \underbrace{u_S(p_{t-}^*) + M_{t-}^*}_{\textit{Sender deviation payoff}}$$

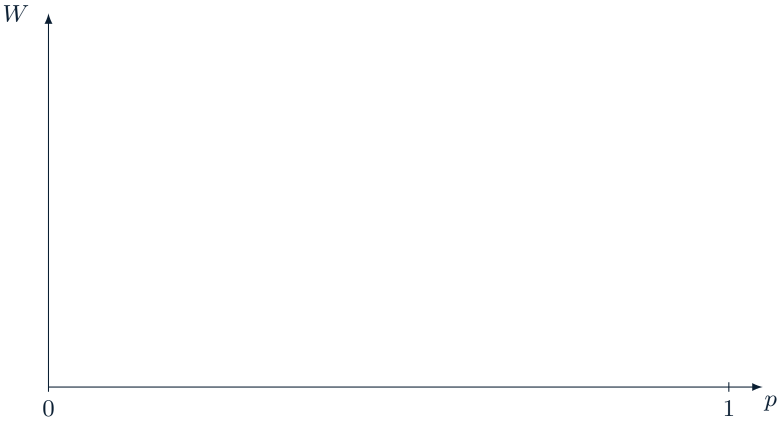
- Receiver incentive compatibility:

$$\mathrm{E}\Big[\underbrace{u_R(p_1^*) - M_1^*}_{\textit{Receiver equilibrium payoff}} \,\Big|\, \mathcal{F}_{t}\Big] \quad \geq \quad \underbrace{u_R(p_t^*) - M_{t-}^*}_{\textit{Receiver deviation payoff}}$$

- If either player deviates from equilibrium, the worst punishment is static autarky.
- Data (information) sharing equilibrium if we restrict the Sender to data (information) sharing strategies.

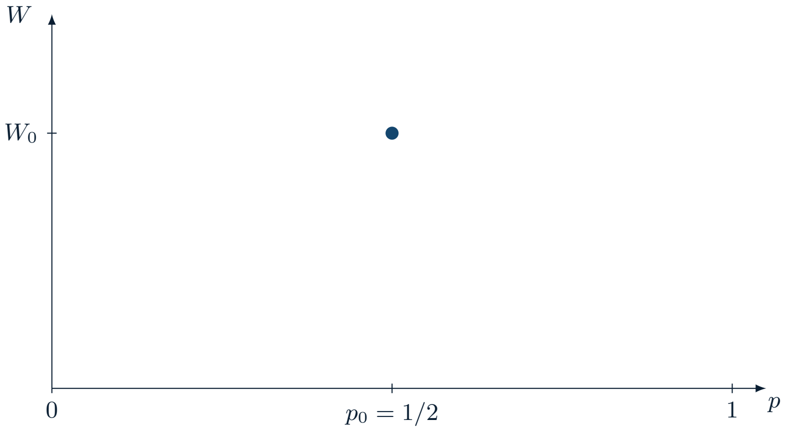
Selling Info without Commitment

Equilibrium Construction/Illustration



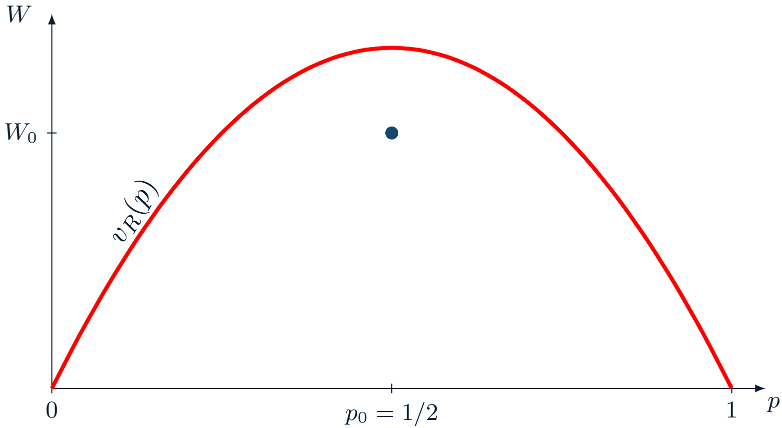
- Posterior $p_t = \mathrm{E}\left[\theta \mid \mathcal{F}_t\right]$ after sharing data up to time t.
- Expected continuation payment $W_t = \mathrm{E}\left[M_1 \mid \mathcal{F}_t\right] - M_{t-}$.

Equilibrium Construction/Illustration



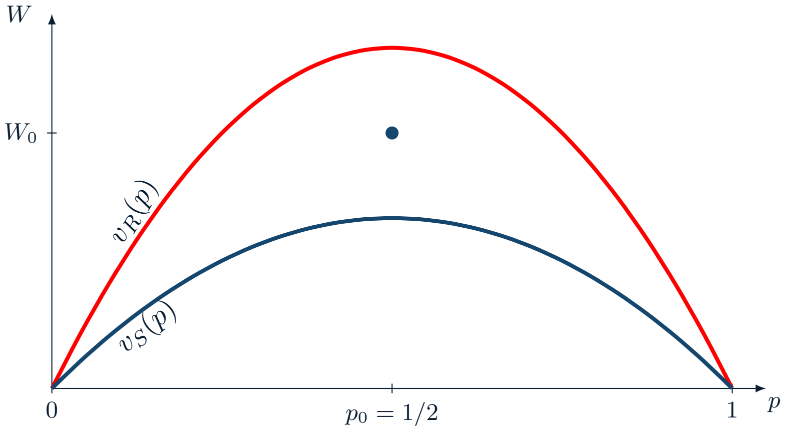
- Start at prior $p_0 = 1/2$ and expected total payment W_0.

Equilibrium Construction/Illustration



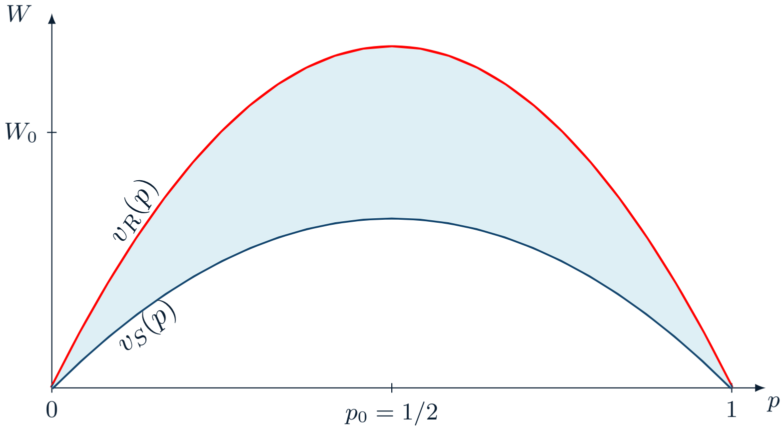
- Receiver's willingness to pay $v_R(p) = \mathrm{E}\left[u_R(\theta)\right] - u_R(p)$.
- Since $v_R(p_0) \geq W_0$, Receiver is ex-ante willing to spend W_0 to learn θ.

Equilibrium Construction/Illustration



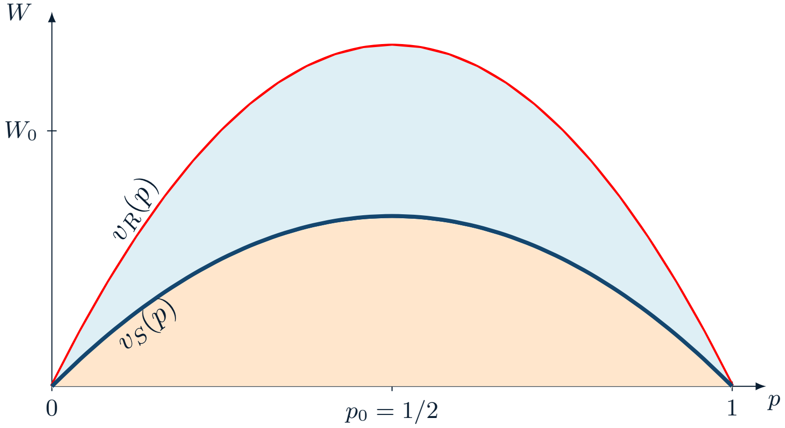
- Sender's cost of sharing $v_S(p) = u_S(p) - \mathrm{E}\left[u_S(\theta)\right]$.
- Since $v_S(p_0) \leq W_0$, Sender is ex-ante willing to share θ in exchange for W_0.

Equilibrium Construction/Illustration



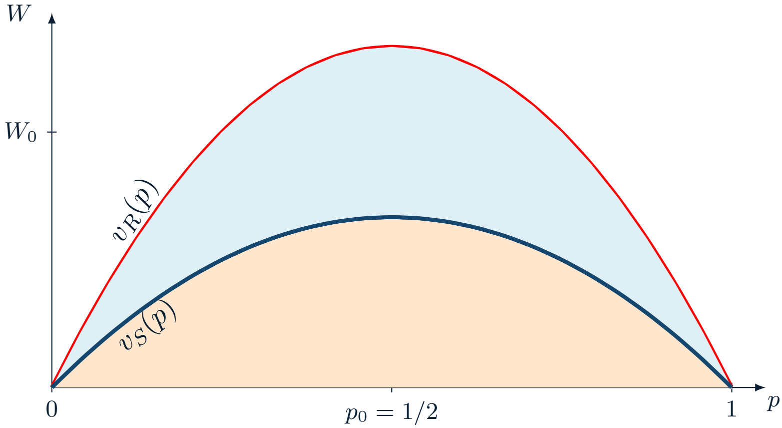
- Ex-ante incentive compatible pairs of (p_0, W_0) at which players are willing to exchange all information about θ for expected payment W_0.

Equilibrium Construction/Illustration



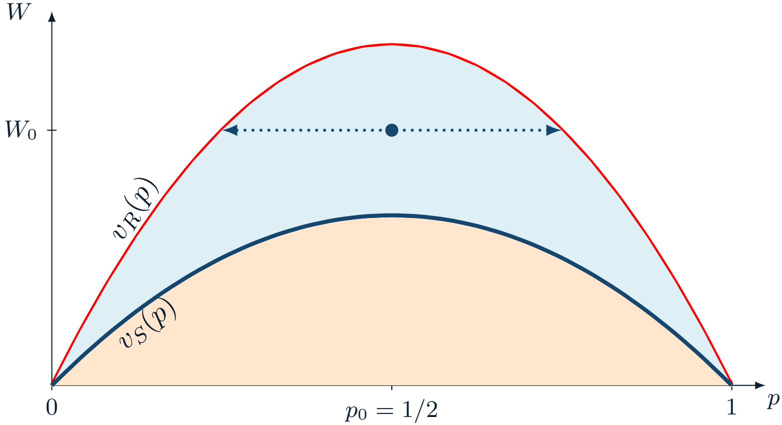
- Below $v_S(p)$, the Sender's continuation cost of information $v_S(p)$ exceeds promised payment W.

Equilibrium Construction/Illustration



- Above $v_R(p)$, the promised payment W exceeds Receiver's benefit from learning information $v_R(p)$.

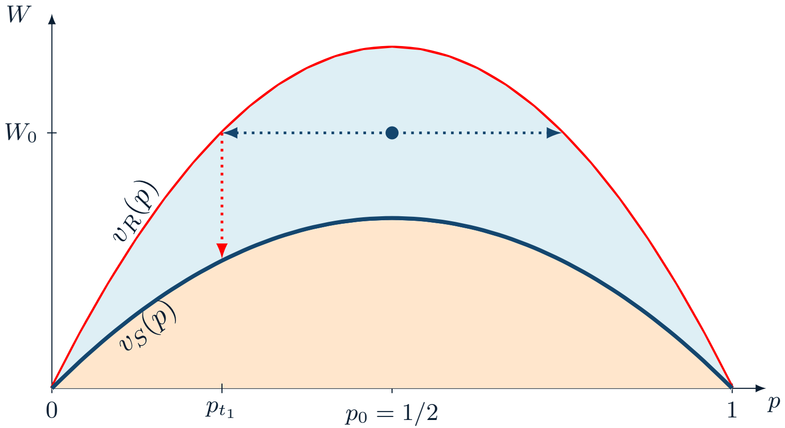
Equilibrium Construction/Illustration



- Equilibrium construction: Sender shares data

$$D_{t_1} \stackrel{def}{=} \inf\{I > 0 : \quad v_R(\mathrm{P}(\theta = 1 | X_I)) = W_0\}\,.$$

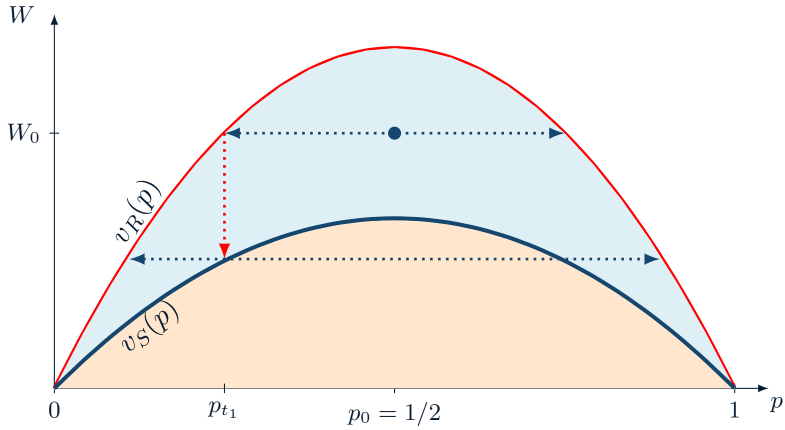
Equilibrium Construction/Illustration



- Once the Receiver observes the realized belief, he pays the Sender

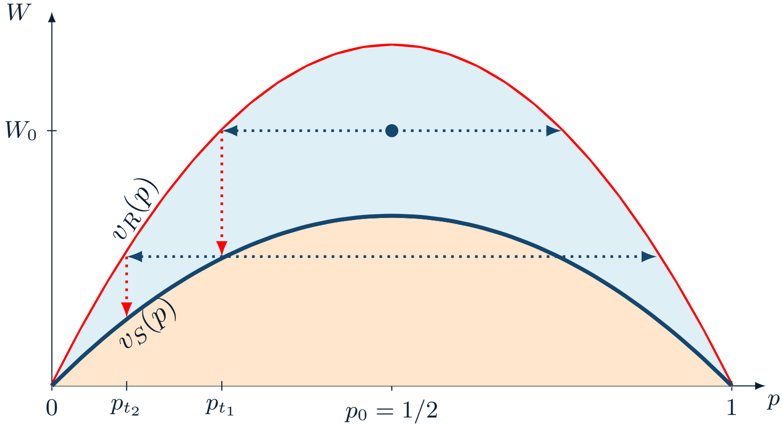
$$M_{t_1} = W_0 - v_S(p_{t_1}).$$

Equilibrium Construction/Illustration



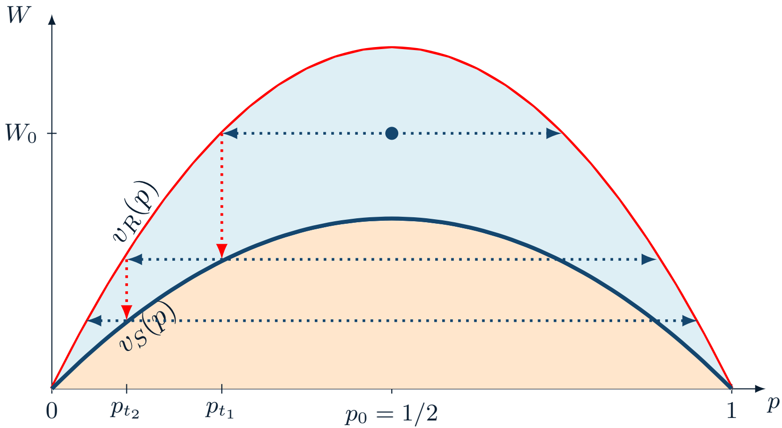
- Players take turns sharing data and making payments.

Equilibrium Construction/Illustration



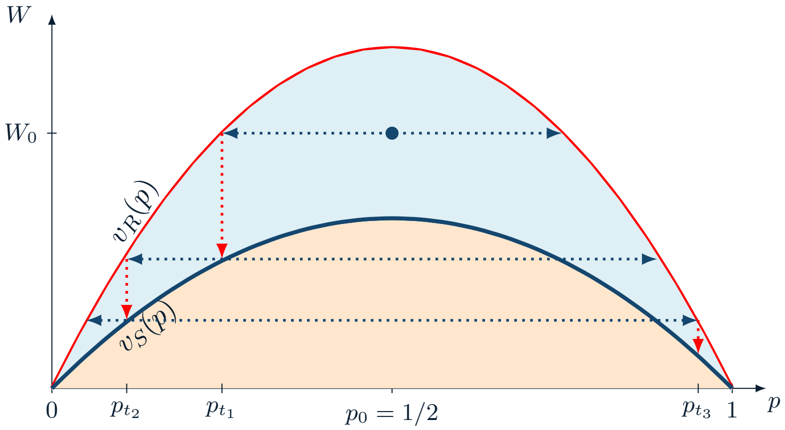
- Every time Sender shares data, he reduces the residual information cost $\Rightarrow$ temptation value declines.

Equilibrium Construction/Illustration



- Every time Receiver makes payment, he reduces his residual budget $\Rightarrow$ his own temptation value declines.

Equilibrium Construction/Illustration



- Sender's residual information cost declines in lockstep with Receiver's residual budget.

Efficiency of Selling Information

Proposition: (Efficient Information Sharing)

There exists an efficient information-sharing equilibrium if and only if the Receiver's budget exceeds the Sender's ex-ante information cost: $\bar{M} \geq v_S(p_0)$.

- Static ex-ante feasibility of full information sharing is sufficient for dynamic implementation.

Efficiency of Selling Information

Proposition: (Efficient Information Sharing)

There exists an efficient information-sharing equilibrium if and only if the Receiver's budget exceeds the Sender's ex-ante information cost: $\bar{M} \geq v_S(p_0)$.

- Static ex-ante feasibility of full information sharing is sufficient for dynamic implementation.
- Intuition: each block can be designed in a way to gradually reduce the residual uncertainty (and thus cost) of future information.
  - Each block must be Goldilocks informative ex-post.
  - Otherwise:
    - extremely informative blocks lead to Receiver hold-up;
    - blocks that increase uncertainty may lead to Sender's hold-up.

Efficiency of Selling Information

Proposition: (Efficient Information Sharing)

There exists an efficient information-sharing equilibrium if and only if the Receiver's budget exceeds the Sender's ex-ante information cost: $\bar{M} \geq v_S(p_0)$.

- Static ex-ante feasibility of full information sharing is sufficient for dynamic implementation.
- Intuition: each block can be designed in a way to gradually reduce the residual uncertainty (and thus cost) of future information.
  - Each block must be Goldilocks informative ex-post.
  - Otherwise:
    - extremely informative blocks lead to Receiver hold-up;
    - blocks that increase uncertainty may lead to Sender's hold-up.
- Information sharing behaves a lot like capital because we are able to generate predictable reductions in residual uncertainty.

Efficiency of Selling Information

Proposition: (Efficient Information Sharing)

There exists an efficient information-sharing equilibrium if and only if the Receiver's budget exceeds the Sender's ex-ante information cost: $\bar{M} \geq v_S(p_0)$.

- Static ex-ante feasibility of full information sharing is sufficient for dynamic implementation.
- Intuition: each block can be designed in a way to gradually reduce the residual uncertainty (and thus cost) of future information.
  - Each block must be Goldilocks informative ex-post.
  - Otherwise:
    - extremely informative blocks lead to Receiver hold-up;
    - blocks that increase uncertainty may lead to Sender's hold-up.
- Information sharing behaves a lot like capital because we are able to generate predictable reductions in residual uncertainty.
- **Contingent blocks are necessary for efficient communication.**

Selling Data without Commitment

Selling Data: Impossibility

Proposition: (Inefficient Data Sharing)

No efficient data-sharing equilibrium exists for any finite budget $\bar{M}$.

Selling Data: Impossibility

Proposition: (Inefficient Data Sharing)

No efficient data-sharing equilibrium exists for any finite budget $\bar{M}$.

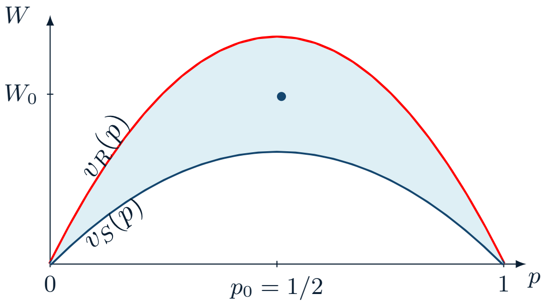
- Sharing data with uncertain informational content creates a risk
  - New data can move the prior excessively in any direction

Selling Data: Impossibility

Proposition: (Inefficient Data Sharing)

No efficient data-sharing equilibrium exists for any finite budget $\bar{M}$.

- Sharing data with uncertain informational content creates a risk
  - New data can move the prior excessively in any direction

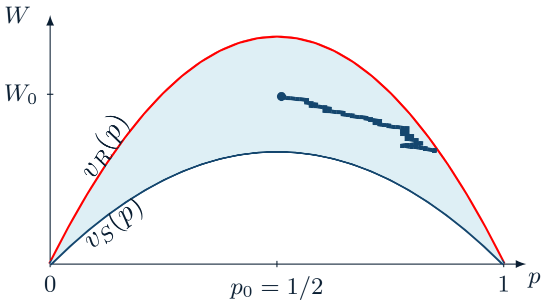


Selling Data: Impossibility

Proposition: (Inefficient Data Sharing)

No efficient data-sharing equilibrium exists for any finite budget $\bar{M}$.

- Sharing data with uncertain informational content creates a risk
  - New data can move the prior excessively in any direction



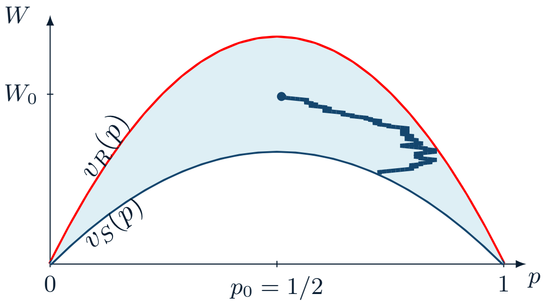
- Initial data can be too informative $\Rightarrow$ Receiver hold-up.
- Receiver's payments lower its temptation value and allow the communication to continue.

Selling Data: Impossibility

Proposition: (Inefficient Data Sharing)

No efficient data-sharing equilibrium exists for any finite budget $\bar{M}$.

- Sharing data with uncertain informational content creates a risk
  - New data can move the prior excessively in any direction



- Subsequent data can be too noisy $\Rightarrow$ Sender hold-up.
- Sender cannot lower its temptation value by sharing more data, and efficient data sharing

Two Simple Extensions

Role of Irreversible Transfers

- Due to noise, new data may increase uncertainty
  - To incentivize the Sender to share data she needs to get paid
- With irreversible transfers the Receiver eventually runs out of budget.
- With reversible transfers Receiver can claw back payments when new data increases uncertainty.

Role of Irreversible Transfers

- Due to noise, new data may increase uncertainty
  - To incentivize the Sender to share data she needs to get paid
- With irreversible transfers the Receiver eventually runs out of budget.
- With reversible transfers Receiver can claw back payments when new data increases uncertainty.

Lemma:

If transfers are reversible, then efficient data-sharing is feasible if and only if $\bar{M} \geq \max_p v_S(p)$.

- With high enough budget $\bar{M}$ efficient data sharing is possible.
- However, ex-ante sufficient budget $\bar{M}$ may be insufficient ex-post.
  - When uncertainty can inrease above the initial $t = 0$ level.

Beyond Binary Fundamentals θ

Variance-Based Preferences

- If θ is binary, then the continuous data sharing induces full support on future posteriors.
  - This creates a risk that cannot be hedged away with any finite budget.

Variance-Based Preferences

- If θ is binary, then the continuous data sharing induces full support on future posteriors.
  - This creates a risk that cannot be hedged away with any finite budget.
- A common framework of quadratic preferences + Gaussian learning does not have this uncertainty.
  - Posterior variance Σ, and thus willingness to pay, decline **deterministically** over time.

Variance-Based Preferences

- If θ is binary, then the continuous data sharing induces full support on future posteriors.
  - This creates a risk that cannot be hedged away with any finite budget.
- A common framework of quadratic preferences + Gaussian learning does not have this uncertainty.
  - Posterior variance Σ, and thus willingness to pay, decline **deterministically** over time.
- Let θ come from an arbitrary distribution $\boldsymbol{F(\cdot)}$.
- Gaussian filter $\boldsymbol{dX_i = \theta\, di + \sigma dB_i}$.

Variance-Based Preferences

- If θ is binary, then the continuous data sharing induces full support on future posteriors.
  - This creates a risk that cannot be hedged away with any finite budget.
- A common framework of quadratic preferences + Gaussian learning does not have this uncertainty.
  - Posterior variance Σ, and thus willingness to pay, decline **deterministically** over time.
- Let θ come from an arbitrary distribution $\boldsymbol{F(\cdot)}$.
- Gaussian filter $\boldsymbol{dX_i = \theta\, di + \sigma dB_i}$.
- Fix variance-based preferences for both players:
  - Receiver: $u_R(\Sigma)$, decreasing in Σ; $v_R(\Sigma) = u_R(0) - u_R(\Sigma)$;
  - Sender: $u_S(\Sigma)$, increasing in Σ; $v_S(\Sigma) = u_S(\Sigma) - u_S(0)$;

Variance-Based Preferences

- If θ is binary, then the continuous data sharing induces full support on future posteriors.
  - This creates a risk that cannot be hedged away with any finite budget.
- A common framework of quadratic preferences + Gaussian learning does not have this uncertainty.
  - Posterior variance Σ, and thus willingness to pay, decline **deterministically** over time.
- Let θ come from an arbitrary distribution $\boldsymbol{F(\cdot)}$.
- Gaussian filter $\boldsymbol{dX_i = \theta\, di + \sigma dB_i}$.
- Fix variance-based preferences for both players:
  - Receiver: $u_R(\Sigma)$, decreasing in Σ; $v_R(\Sigma) = u_R(0) - u_R(\Sigma)$;
  - Sender: $u_S(\Sigma)$, increasing in Σ; $v_S(\Sigma) = u_S(\Sigma) - u_S(0)$;
- Full information is efficient: $u_R(0) + u_S(0) \geq u_R(\Sigma) + u_S(\Sigma)$ for all Σ.

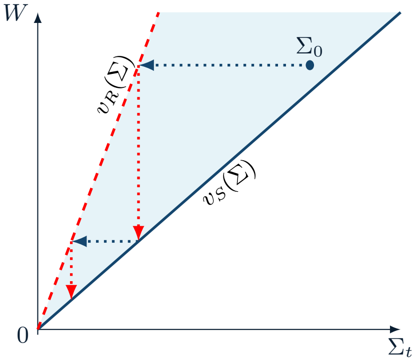
Efficiency of Information Sharing

Proposition: (Information Sharing Efficiency)

There exists an information sharing equilibrium that attains efficiency whenever the Receiver's starting budget exceeds the Sender's static information cost, i.e., $\bar{M} \geq v_S(\Sigma_0)$.

- In the limit as all data is shared, posterior variance declines to 0.
- The size of each block is random, but the posterior variance can be predictable.
- Consider data batches that reduce posterior variance to pre-specified levels.

$$D_{t_k} = \inf \left\{ I : \, v_R \left(\Sigma(I, X_I) \right) = \bar{M} - M_{t_k -} \right\}$$



Normal Priors are Special for Data Sharing

Proposition: (Excess Budget)

Gaussian prior θ is the only family of distributions for which $\bar{M} \geq v_S(\Sigma_0)$ is necessary and sufficient to support an efficient data-sharing equilibrium for any Sender/Receiver preferences that feature gains from trade: $v_R(\Sigma) \geq v_S(\Sigma)$ for every $\Sigma \leq \Sigma_0$.

- If θ is Gaussian, then having just enough budget to compensate the Sender's ex-ante information cost $v_S(\Sigma_0)$ is sufficient.
  - Standard result. Data behaves just like capital.

Normal Priors are Special for Data Sharing

Proposition: (Excess Budget)

Gaussian prior θ is the only family of distributions for which $\bar{M} \geq v_S(\Sigma_0)$ is necessary and sufficient to support an efficient data-sharing equilibrium for any Sender/Receiver preferences that feature gains from trade: $v_R(\Sigma) \geq v_S(\Sigma)$ for every $\Sigma \leq \Sigma_0$.

- If θ is Gaussian, then having just enough budget to compensate the Sender's ex-ante information cost $v_S(\Sigma_0)$ is sufficient.
  - Standard result. Data behaves just like capital.
- If θ is non-Gaussian, then efficient data-sharing requires excess budget $\bar{M}$ to absorb dynamic uncertainty.
  - Distributions "closer" to Gaussian require less slack.
  - Distributions similar to binary, e.g. discrete, will require "infinite" slack.

Normal Priors are Special for Data Sharing

Proposition: (Excess Budget)

Gaussian prior θ is the only family of distributions for which $\bar{M} \geq v_S(\Sigma_0)$ is necessary and sufficient to support an efficient data-sharing equilibrium for any Sender/Receiver preferences that feature gains from trade: $v_R(\Sigma) \geq v_S(\Sigma)$ for every $\Sigma \leq \Sigma_0$.

- If θ is Gaussian, then having just enough budget to compensate the Sender's ex-ante information cost $v_S(\Sigma_0)$ is sufficient.
  - Standard result. Data behaves just like capital.
- If θ is non-Gaussian, then efficient data-sharing requires excess budget $\bar{M}$ to absorb dynamic uncertainty.
  - Distributions "closer" to Gaussian require less slack.
  - Distributions similar to binary, e.g. discrete, will require "infinite" slack.
- The magnitude of the excess budget depends on the magnitude of gains from information exchange.

Gradualism and Information: Takeaway

- Limited commitment frictions require gradualism.
- Gradual information revelation creates short-term uncertainty risks.
- Information design effectively hedges such uncertainty risks.
- Granular data sharing can lead to inefficiencies.
  - Efficiency is impossible for discrete fundamentals.
  - Excess budget is needed for dynamic implementation for all non-Gaussian fundamentals.
  - The magnitude of additional financial flexibility is decreasing in the gains from information trade.

Gradualism and Information: Takeaway

- Limited commitment frictions require gradualism.
- Gradual information revelation creates short-term uncertainty risks.
- Information design effectively hedges such uncertainty risks.
- Granular data sharing can lead to inefficiencies.
  - Efficiency is impossible for discrete fundamentals.
  - Excess budget is needed for dynamic implementation for all non-Gaussian fundamentals.
  - The magnitude of additional financial flexibility is decreasing in the gains from information trade.
- Thank you!